Preliminary Dense-RHS CoRL Compact Results

Autogenerated from local TD-MPC2-JAX run artifacts

Scope. This preliminary report summarizes the completed compact campaign "dense_rhs_corl_compact_v2" using local metric caches refreshed from the NCC run directories. It is intended as an internal CoRL-style evidence package, not as the final v3 matrix.

Headline. Across 33 matched valid seed pairs, Adaptive RHS changes final score by **4.72%** $\pm$ **23.26%** relative to the fixed TD-MPC2 paper horizon, with an RHS win rate of **66.67%**. The summary excludes 3 missing, nonfinite, or unmatched pairs.

Experimental Setup

We compare the TD-MPC2 fixed paper horizon against Adaptive RHS under matched environment, regime, and seed cells. Environments are quadruped-run, cheetah-run, hopper-hop, finger-turn_hard, fish-swim, cartpole-swingup; regimes are clean and chaos; seeds are 1, 7, 15; training budget is 500000 environment steps. All rows come from `experiments/corl_compact_v2_ledger.csv`; each row records the SLURM job id, git commit, remote run directory, score, runtime, and checkpoint status.

Aggregate Result

Scope	n	RHS delta (%)	RHS wins	Runtime ratio
all valid pairs	33	4.72 $\pm$ 23.26	66.67%	1.13
chaos	15	11.43 $\pm$ 31.37	73.33%	1.13
clean	18	-0.87 $\pm$ 11.64	61.11%	1.12

Learning Curves

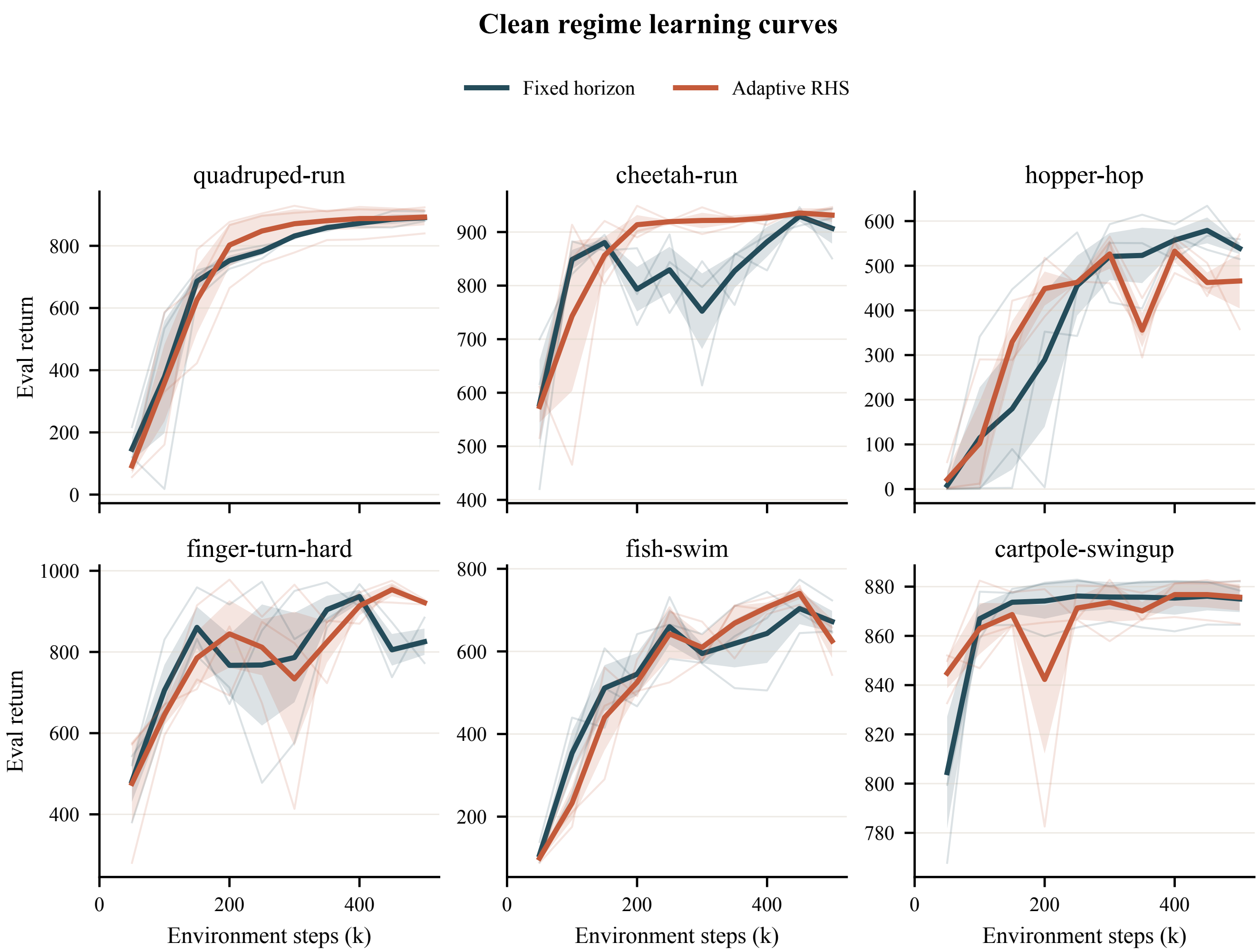

Figure 1: Clean-regime evaluation return. Thin traces are individual seeds; bold traces show the seed mean with standard error.

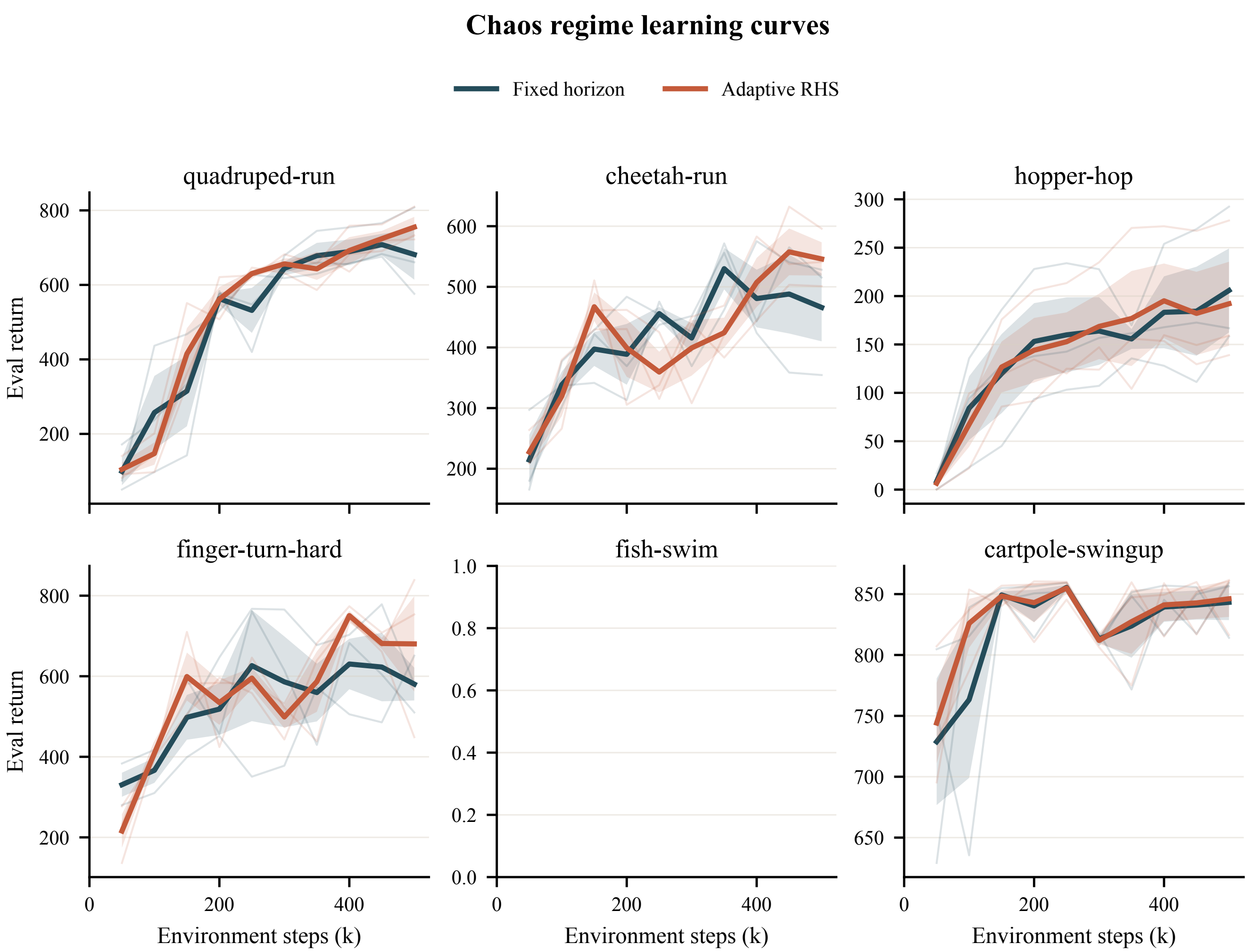

Figure 2: Chaos-regime evaluation return under matched randomized/noisy evaluation.

Matched Percent Delta

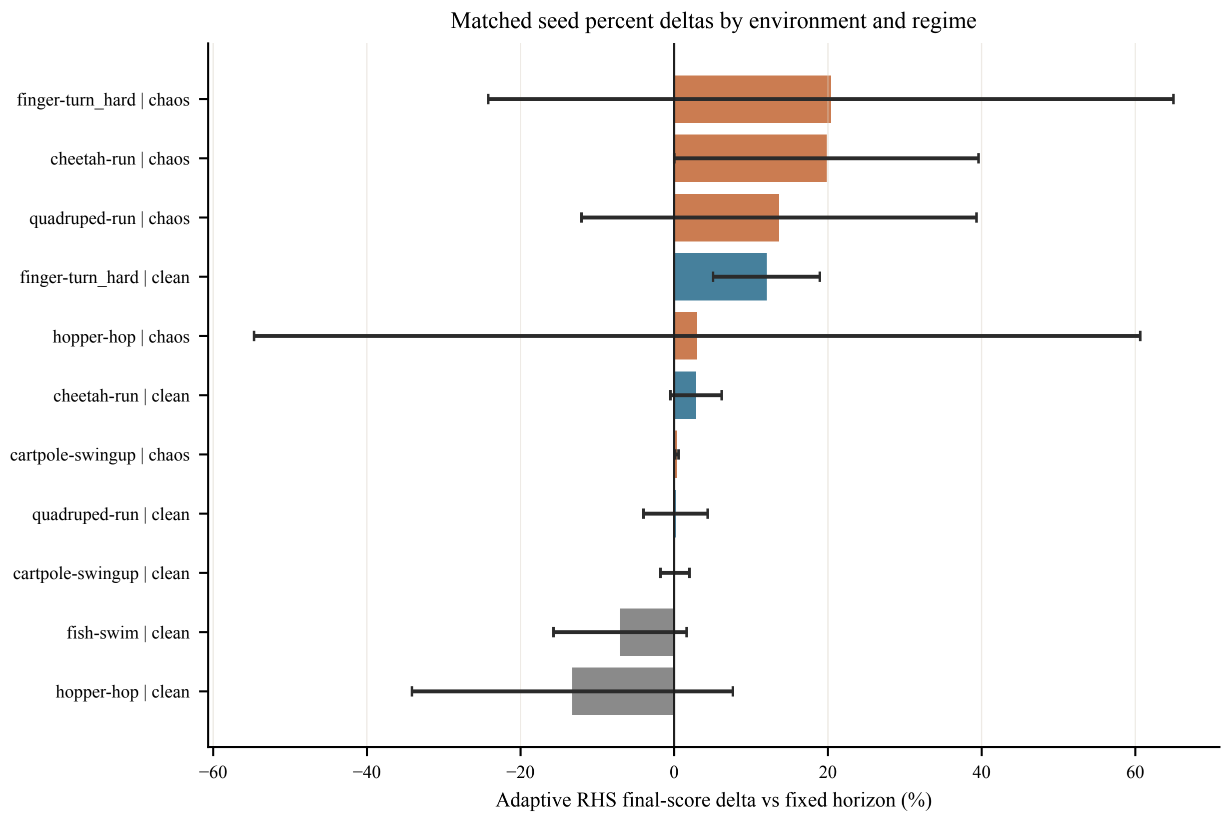

Figure 3: Per-environment and per-regime Adaptive RHS percent delta against the matched fixed-horizon baseline. Error bars are across seeds.

Environment	Regime	n	RHS delta (%)	RHS wins	RHS h
cartpole-swingup	chaos	3	0.34 ± 0.22	100.0%	2.12
cartpole-swingup	clean	3	0.09 ± 1.89	66.67%	1.81
cheetah-run	chaos	3	19.80 ± 19.80	100.0%	2.22
cheetah-run	clean	3	2.84 ± 3.34	66.67%	1.63
finger-turn_hard	chaos	3	20.38 ± 44.59	66.67%	2.11
finger-turn_hard	clean	3	11.99 ± 6.95	100.0%	1.62
fish-swim	clean	3	-7.05 ± 8.67	33.33%	1.52
hopper-hop	chaos	3	2.98 ± 57.66	33.33%	2.53
hopper-hop	clean	3	-13.25 ± 20.88	33.33%	1.85
quadruped-run	chaos	3	13.63 ± 25.71	66.67%	2.59
quadruped-run	clean	3	0.18 ± 4.18	66.67%	2.16

Efficiency and Pareto View

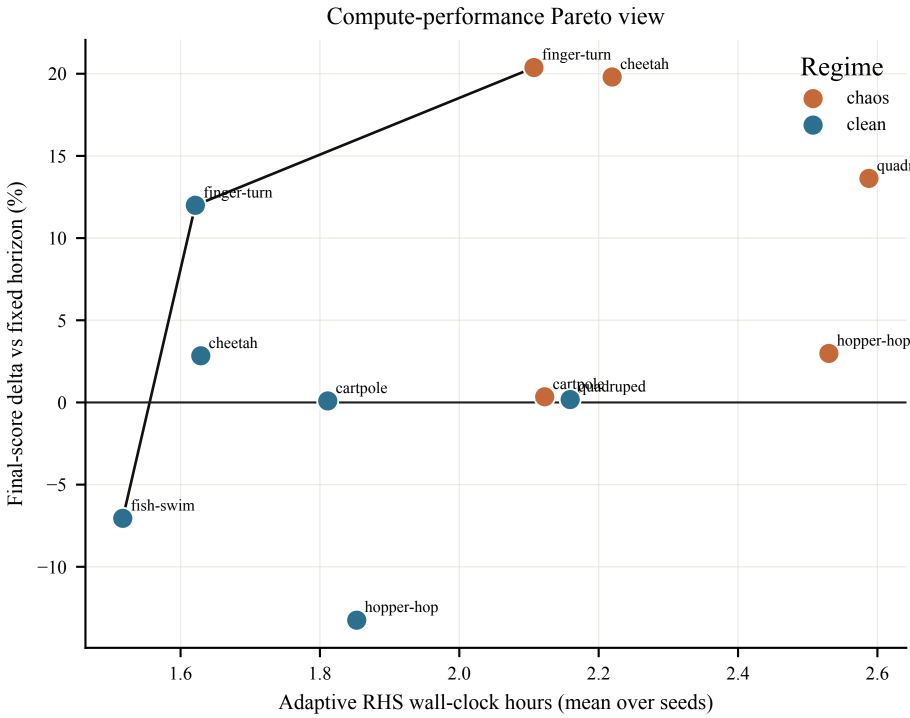

Figure 4: Compute-performance Pareto view. The x-axis is mean Adaptive RHS wall-clock hours and the y-axis is normalized final-score delta versus the matched fixed-horizon baseline; the black line highlights nondominated points.

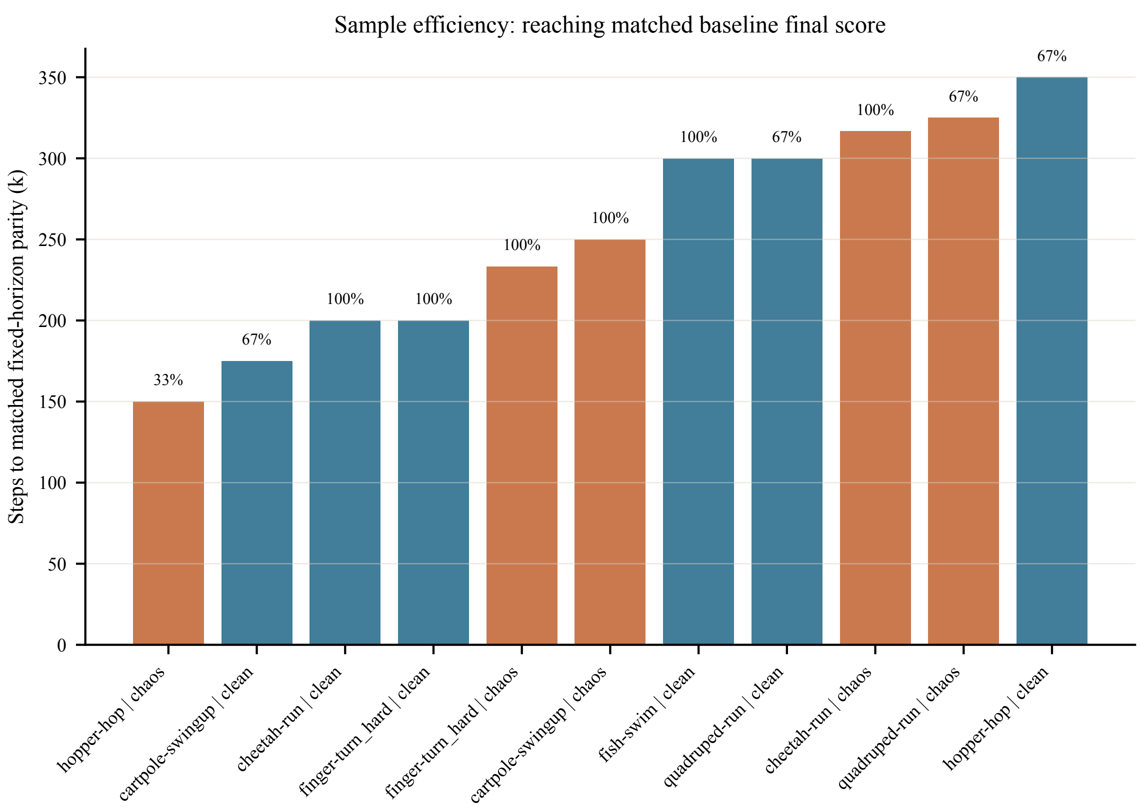

Figure 5: Sample efficiency, measured as the first Adaptive RHS evaluation point that reaches the matched fixed-horizon final score. Labels above bars indicate the fraction of seeds reaching parity.

RHS Diagnostics

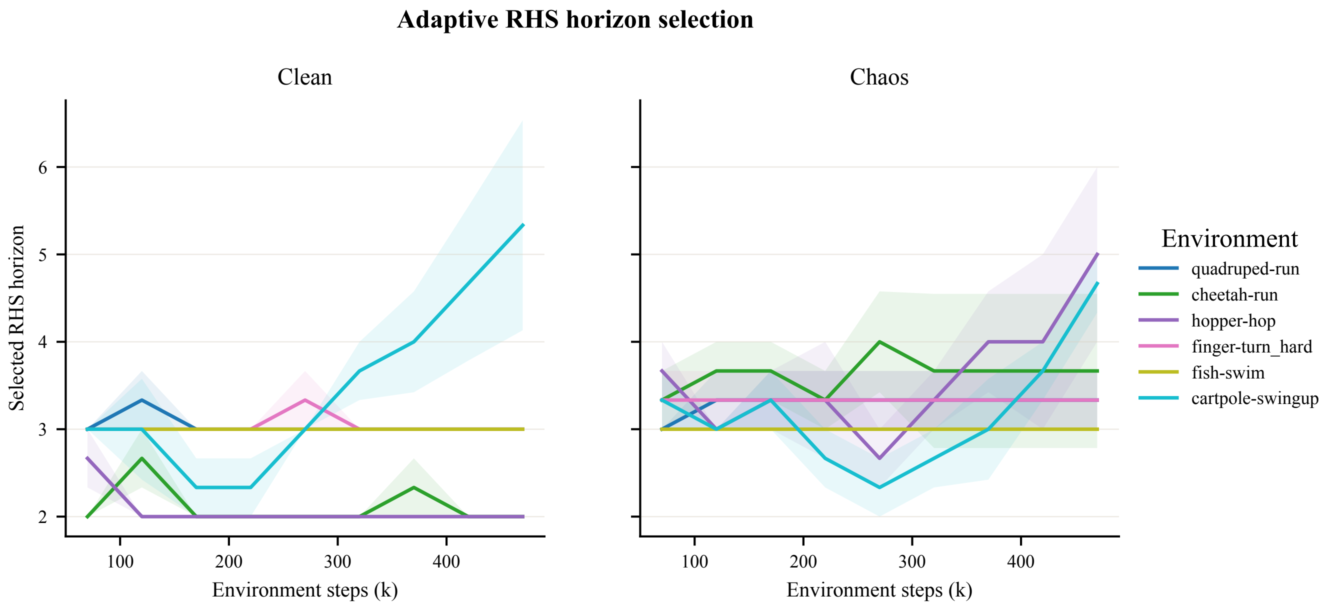

Figure 6: Adaptive RHS selected horizon over training, averaged across seeds.

Limitations

This is a preliminary report for the completed compact run (72 terminal rows). It includes the v2 environment set, including fish-swim. Fish was later removed from the v3 final matrix after isolated MJX chaos-push diagnostics exposed nonfinite state/reward failures, so this report should be framed as preliminary evidence rather than the final CoRL table. The final v3 package should reuse the same report template with the updated environment set and six seeds.

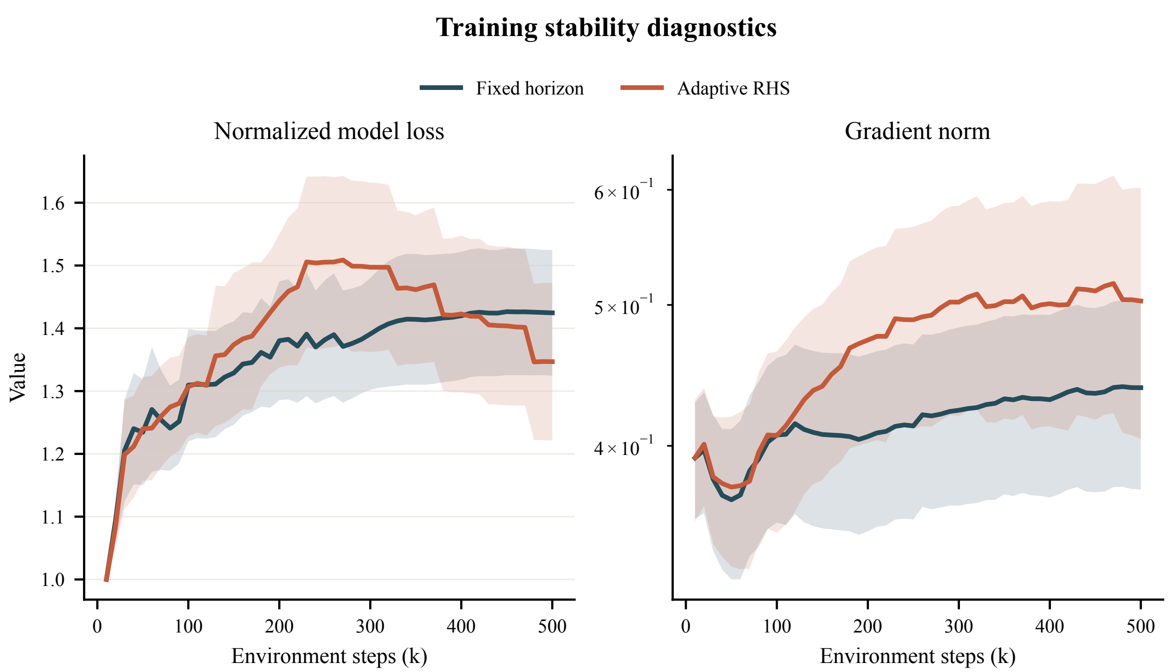

Figure 7: Training stability diagnostics pooled across runs. Total loss is normalized per run by its first finite value; gradient norms use a log axis.